LASSO Regression

LASSO Regression will teach us how to build models when we have many potential predictors but expect that only some of them contain useful information. Unlike Ridge Regression, which generally shrinks all coefficients without eliminating variables, LASSO uses an L1 penalty that can shrink selected coefficients exactly to zero. It performs regularisation and automatic feature selection simultaneously, allowing us to construct simpler and more interpretable predictive models while controlling overfitting.

Problem	Title	Research Anchor	Dataset
Problem 11	Forecasting U.S. Inflation	Stock and Watson's inflation-forecasting framework; LASSO methodology from Tibshirani	FRED monthly U.S. macroeconomic data
Problem 12	Predicting the U.S. Equity Premium	Welch and Goyal's equity-premium prediction framework; LASSO methodology from Tibshirani	Amit Goyal Data Library: U.S. equity-premium and economic predictor dataset
Problem 13	Predicting GDP Growth Under Climate Stress	Burke, Hsiang & Miguel's nonlinear relationship between annual temperature & economic productivity; LASSO methodology from Tibshirani	Copernicus Climate Change Service ERA5 reanalysis + OWID CO_2 & Greenhouse Gas Emissions data

Table 1: *Overview of Problems, Research Anchors, and Datasets*

11. US Inflation Forecast Using Macroeconomic Indicators

Problem 11: Forecasting U.S. Inflation Using Macroeconomic Indicators with LASSO

Research Anchor 1: Stock, J. H. & Watson, M. W. (1999), "Forecasting Inflation", Journal of Monetary Economics, 44(2), 293–335.

Research Anchor 2: Tibshirani, R. (1996), "Regression Shrinkage and Selection via the Lasso".

Forecasting inflation is one of the classic problems in macroeconomics. Central banks, governments, financial institutions and investors want to know whether current economic conditions contain information about future inflation.

Stock and Watson examined whether a large collection of economic indicators could improve U.S. inflation forecasts relative to conventional Phillips-curve models. Their analysis particularly emphasised out-of-sample forecasting.

Our exercise is **not an exact reproduction of their paper**. Instead, it is a simplified replication-style exercise inspired by their economic question, combined with a modern variable-selection method.

Tibshirani introduced LASSO as a regression method that penalises the absolute magnitudes of coefficients. A distinctive feature of this penalty is that some coefficients can become exactly zero, thereby simultaneously performing regression and variable selection.

Question: *Which macroeconomic indicators contain the most useful information for forecasting U.S. inflation twelve months ahead, and can LASSO eliminate indicators that contribute little predictive information?*

Results and Interpretation

Cross-validation selected an optimal LASSO regularisation parameter of **0.2719**. Of the eight candidate macroeconomic predictors, the model retained only the **Federal Funds Rate**, while current inflation, unemployment, industrial-production growth, money-supply growth, oil-price growth, the term spread and housing-start growth were all shrunk to zero.

The Federal Funds Rate received a standardised LASSO coefficient of **0.5085**, compared with an OLS coefficient of **1.2530**. I find that the L1 penalty substantially reduces the magnitude of the interest-rate coefficient while eliminating the remaining predictors completely.

I interpret this as evidence that the Federal Funds Rate provided the most robust predictive information under the penalty chosen through time-series cross-validation.

However, I do not interpret the positive coefficient causally. Higher policy rates do not imply that monetary tightening causes higher inflation. The Federal Reserve typically increases interest rates in response to existing or expected inflationary pressure, meaning that high policy rates can coincide with inflation remaining elevated over the subsequent twelve months.

The forecasting results show an important trade-off between model simplicity and predictive performance.

LASSO achieves an RMSE of **2.5638** and an MAE of **1.6691**, compared with **3.3681** and **2.1979** respectively for OLS. Regularisation produces a substantial improvement over the unrestricted linear regression.

In particular, the LASSO RMSE is approximately 24% lower than the OLS RMSE, suggesting that the unrestricted OLS specification contains relationships that do not generalise well to the out-of-sample period.

The comparison with the naive inflation-persistence model is more nuanced. The naive model records the lowest RMSE at **2.5095**, slightly outperforming LASSO, whereas LASSO records the lowest MAE at **1.6691**, compared with **1.8298** for the naive model.

I find that LASSO produces smaller forecast errors in a typical month, but its large errors during the post-pandemic inflation surge are penalised heavily by RMSE. Consequently, **no single model dominates according to every evaluation metric.**

All three models generate negative out-of-sample (R^2) values. The naive, OLS and LASSO models produce (R^2) values of **-0.4437, -1.6007 and -0.5070**, respectively. This indicates that none of the models successfully captures the exceptional inflation dynamics of the test period relative to the statistical mean benchmark. The particularly poor OLS result reinforces the value of regularisation, although LASSO itself remains unable to forecast the major structural shift in inflation.

The actual-versus-forecast figure shows that LASSO generates relatively smooth inflation forecasts, generally between approximately 2% and 3%. It fails to anticipate the dramatic rise in inflation during 2020–2022, when actual inflation eventually exceeds 8%. This behaviour is consistent with the model selecting only the Federal Funds Rate. Policy rates remained very low during the initial inflation surge and increased substantially only after inflation was already elevated. The **resulting model responds too slowly to the inflation shock.**

The comparison across all three forecasting models further illustrates their differences.

- The naive model follows inflation persistence and consequently captures much of the magnitude of the inflation increase, although its peak occurs later than the actual peak.
- OLS is considerably less stable and even generates very low or negative inflation forecasts during part of the period in which actual inflation is rising sharply.

- LASSO produces the smoothest forecasts and avoids the extreme OLS predictions, but this stability comes at the cost of failing to capture the magnitude of the inflation shock.

The LASSO coefficient-path figure demonstrates the mechanics of L1 regularisation. At very small values of alpha, most predictors retain non-zero coefficients. As alpha increases, the coefficients progressively move toward zero. Most variables disappear relatively quickly, while the Federal Funds Rate survives over a much wider range of penalty values. At the cross-validation-selected alpha of **0.2719**, only the Federal Funds Rate remains. This demonstrates **how increasing regularisation transforms a relatively complex macroeconomic model into a sparse specification.**

The feature-selection plot provides the same result from another perspective. At very weak regularisation, all eight predictors are retained. As the penalty becomes stronger, the number of active predictors falls until only one variable remains around the selected alpha. At still larger alpha values, every predictor is eliminated and the **model effectively reduces to an intercept-only forecast.**

Finally, the residual diagnostics reveal an important limitation of the model. Forecast errors are strongly positive during the 2020–2022 inflation surge, reaching approximately six to seven percentage points at their largest. Because the residual is defined as actual minus predicted inflation, **these positive errors show severe underprediction.**

The residuals also occur in persistent clusters rather than appearing randomly around zero, suggesting that the relationship between the selected predictors and inflation changes over time. The predicted-value-versus-residual plot confirms that the **model performs particularly poorly during the exceptional inflation regime.**

(Refer to Appendix A.11 for more on Results)

Conclusion

Overall, I find that **LASSO successfully demonstrates its principal advantage over OLS**: it can produce a substantially more parsimonious model while improving forecasting performance relative to an unrestricted regression. However, the exercise also demonstrates that aggressive feature selection does not guarantee superior forecasts.

The naive model narrowly achieves a better RMSE during the test period, and LASSO fails to capture the post-pandemic inflation shock. An especially important finding is that LASSO eliminates current inflation even though inflation persistence performs relatively well during the test period. This illustrates that **variables selected using historical cross-validation need not remain optimal when the economic data-generating process undergoes a structural change.**

The results demonstrate both the usefulness and the limitations of LASSO for macroeconomic forecasting: regularisation can control overfitting and create interpretable sparse models, but it cannot protect a forecasting model against major structural breaks or ensure that historically selected predictors remain optimal in a new economic regime.

12. Predicting the U.S. Equity Premium

Problem 12: Predicting the U.S. Equity Premium Using Economic and Financial Predictors

Research Anchor 1: Welch, I. and Goyal, A. (2008), "A Comprehensive Look at the Empirical Performance of Equity Premium Prediction", Review of Financial Studies.

Research Anchor 2: Tibshirani (1996), "Regression Shrinkage and Selection via the Lasso".

Question: *Which financial and macroeconomic variables are selected by LASSO when forecasting the one-month-ahead U.S. equity premium, and does the resulting sparse model outperform OLS and the historical-average benchmark out of sample?*

Researchers have proposed many predictors: dividend yields, earnings ratios, interest rates, term spreads, default spreads, inflation, stock-market volatility and measures of equity issuance.

Welch and Goyal (2008) conducted a comprehensive investigation of these predictors and found that many apparently successful relationships performed poorly when evaluated genuinely out of sample.

Problem 12 will have 14 candidate predictors, many of which are economically related to one another.

Variable	Meaning	Variable	Meaning
dp	Dividend-price ratio	svar	Stock-market variance
dy	Dividend yield	bm	Book-to-market ratio
ep	Earnings-price ratio	ntis	Net equity issuance
de	Dividend-payout ratio	tbl	Treasury-bill rate

lty	Long-term government-bond yield	dfy	Default yield spread
ltr	Long-term government-bond return	dfr	Default return spread
tms	Term spread	infl	Inflation

Results and Interpretation

Time-series cross-validation selected an optimal LASSO regularisation parameter of **0.7229**. The most striking result is that LASSO eliminated **all fourteen candidate predictors**, assigning a coefficient of zero to every valuation, interest-rate, credit-market, volatility, issuance and inflation variable included in the model. Consequently, the final LASSO specification contains only an intercept and produces effectively the same forecast as the historical-mean equity-premium benchmark.

This result should not be interpreted as evidence that the fourteen variables have no economic relevance. Rather, I find that, conditional on the forecasting design and the penalty selected through time-series cross-validation, none of them provides sufficiently robust incremental information for forecasting the next month's U.S. equity premium.

This distinction is particularly important because OLS assigns non-zero coefficients to every predictor. For example, the standardised OLS coefficients include **1.8110 for the dividend-price ratio, -2.1541 for dividend yield and 1.3253 for the earnings-price ratio**. The contrast with LASSO demonstrates that sizeable in-sample regression coefficients do not necessarily translate into reliable out-of-sample predictive information.

The forecast-performance results reinforce this conclusion. The historical-mean benchmark achieves an RMSE of **4.0730** and an MAE of **3.1368**. LASSO produces

exactly the same values because its complete elimination of the predictor set reduces the model to an intercept-only forecast. OLS performs worse, with an RMSE of **4.3864** and an MAE of **3.2078**. Its RMSE is approximately **7.7% higher** than that of the historical-mean benchmark. Despite incorporating fourteen economic and financial predictors, the unrestricted linear regression does not improve out-of-sample forecasting performance.

The equity-premium-specific out-of-sample (R^2) provides an especially clear comparison. OLS produces an (R^2 (OOS)) of **-0.1598**, indicating that its squared forecast errors are approximately 15.98% greater than those of the historical-mean benchmark. LASSO produces an (R^2 (OOS)) of **0.0000** because its forecasts are identical to the benchmark.

I find no evidence that either specification improves upon the historical mean, although LASSO successfully prevents the deterioration in forecasting performance generated by unrestricted OLS.

The standard test-sample (R^2) values are **-0.0164 for both the historical mean and LASSO and -0.1789 for OLS**. These values differ conceptually from the equity-premium (R^2_{OOS}). The standard (R^2) uses the mean realised equity premium within the test sample as its reference, whereas (R^2 (OOS)) directly compares each predictive model with the historical-mean forecast constructed from the training sample. LASSO has an (R^2 (OOS)) of zero even though its conventional test-sample (R^2) is slightly negative.

The direction-accuracy results also require careful interpretation. The historical mean and LASSO both achieve **68.13%**, compared with **63.19% for OLS**. Because the historical-average equity premium is positive, both the historical-mean model and the intercept-only LASSO specification forecast a positive equity premium every month. The 68.13% accuracy largely reflects the fact that realised equity premiums are positive

in approximately 68% of test-sample months rather than demonstrating an ability to time changes in the direction of the market.

The actual-versus-forecast figure illustrates the difficulty of monthly equity-premium prediction. Realised monthly excess returns fluctuate sharply, including several movements of approximately plus or minus 10%, while the historical-mean and LASSO forecasts remain essentially constant.

OLS generates substantially more variation in its forecasts, but this additional flexibility does not translate into superior accuracy. The result suggests that most short-horizon variation in realised equity returns behaves as noise relative to the information contained in the selected predictor set.

The LASSO coefficient-path figure demonstrates how the final sparse specification emerges. At very small penalty values, many variables receive sizeable positive or negative coefficients. As alpha increases, these coefficients progressively shrink toward zero. At the cross-validation-selected alpha of **0.7229**, the final model assigns zero coefficients to all fourteen predictors. Cross-validation favours strong regularisation because the reduction in estimation variance outweighs the predictive information contributed by the individual regressors.

This conclusion is also visible in the cross-validation-error curve.

Forecasting RMSE declines materially as the regularisation penalty increases from very small values and reaches its minimum near the selected alpha. The result indicates that models containing relatively unrestricted predictor coefficients generalise poorly to later observations, whereas increasing shrinkage improves cross-validated forecasting performance until an extremely sparse specification is reached.

The cumulative squared-error comparison provides additional insight into the instability of OLS forecasting performance. During much of the period from approximately 2010 to 2019, OLS accumulates positive forecasting gains relative to the historical mean, suggesting that the predictor-based model initially performs well.

Around 2020, however, its cumulative advantage collapses sharply and becomes strongly negative. Subsequent observations do not reverse this deterioration, and the cumulative forecasting loss continues to increase through the remainder of the sample. This demonstrates that an apparent forecasting relationship can perform successfully for an extended period yet fail dramatically when market conditions change.

The LASSO cumulative-gain line remains at zero throughout the test sample because the model generates the same forecasts as the historical-mean benchmark. Although LASSO does not create positive forecasting gains, regularisation protects the model from the substantial losses experienced by OLS.

(Refer to Appendix A.12 for more on Results)

Conclusion

Overall, my results are consistent with the central difficulty highlighted in the equity-premium forecasting literature: economically plausible predictors do not necessarily generate forecasts that outperform a simple historical-average benchmark. LASSO responds to this instability by eliminating the entire predictor set rather than retaining relationships that fail to generalise reliably. In this application, the value of regularisation lies not in producing superior forecasts to the historical mean, but in recognising that the available predictors provide insufficient robust information and preventing the model from overfitting noisy return relationships.

The exercise also highlights an important distinction between prediction and economic explanation. A zero LASSO coefficient does not establish that a variable has no relationship with expected returns or no economic significance. It indicates only that the variable does not contribute enough stable incremental forecasting information to survive the chosen L1 penalty within this particular model and sample.

Note: *Our historical-mean benchmark is the fixed mean estimated from the training sample. The original Welch–Goyal forecasting literature commonly evaluates forecasts against a*

recursively updated historical-average benchmark. So, this remains a replication-extension inspired by their forecasting framework, rather than claiming to reproduce their exact empirical procedure.

13. Predicting GDP Growth

Problem 13: Predicting GDP Growth Under Climate Stress Using Economic, Energy and Temperature Indicators

Research Anchor 1: Burke, M., Hsiang, S.M. and Miguel, E. (2015) 'Global non-linear effect of temperature on economic production', *Nature*, 527, pp. 235–239.

Research Anchor 2: Tibshirani (1996), "Regression Shrinkage and Selection via the Lasso".

Dataset 1 — Climate data: Annual average surface temperature by country.

The underlying temperature series is derived from the Copernicus Climate Change Service ERA5 reanalysis. OWID converts it into country-level, latitude-weighted annual temperature estimates.

Dataset 2 — Economic, energy and emissions data

The OWID CO_2 and Greenhouse Gas Emissions dataset provides annual country-level observations and incorporates GDP, population, energy use and emissions information from several underlying sources.

We are **not exactly reproducing their causal specification**. Instead, we will construct a LASSO-based prediction extension: *Climate Economics literature + Macroeconomic predictors + Energy/Emission Indicators + LASSO feature selection*.

Can LASSO identify climate, energy and economic variables that contain useful information for forecasting next-year real GDP-per-capita growth across countries?

Does adding direct temperature information improve forecasting performance relative to a model containing only economic and energy indicators?

Problem 13 introduces: ***Country x Year panel data***

We will use 14 predictors.

- temperature, temperature_squared temperature_change heat_stress
- current_gdp_pc_growth, log_gdp_pc , population_growth
- co2_growth, co2_per_capita, co2_per_gdp, energy_per_capita , energy_per_gdp, coal_co2_per_capita, oil_co2_per_capita

Results and Interpretation

The final modelling sample contains **3,628 country-year observations across 122 countries**, covering predictor years from 1991 to 2021. I use observations from 1991 to 2013 for model estimation and reserve 2015 to 2021 for out-of-sample testing, leaving a one-year gap because the dependent variable represents next-year real GDP-per-capita growth. The average GDP-per-capita growth rate in the training sample is **2.98%**.

Time-series cross-validation selected an optimal alpha of **0.0070** for the full Climate-Economy LASSO model.

This represents relatively weak regularisation. Of the 14 candidate predictors, LASSO retains **13** and eliminates only **temperature-squared**. The result demonstrates that LASSO does not necessarily produce an extremely sparse model. In this application, cross-validation indicates that most predictors contain sufficient information to justify retaining them.

Three of the four direct climate variables survive the LASSO penalty:

- annual average temperature
- annual temperature change
- the heat-stress indicator

Their standardised coefficients are -0.4444, 0.2265 and 0.6364, respectively, while the temperature-squared coefficient is reduced to zero.

I do not interpret these coefficients causally because the climate variables are closely related by construction and because the model is designed for prediction rather than causal identification. Instead, their retention indicates that temperature information contained predictive signal within the training data conditional on the remaining economic and energy indicators.

The elimination of temperature squared is also notable. Rather than retaining the quadratic temperature specification directly, LASSO retains temperature together with the threshold-based heat-stress variable. The selected specification resembles a piecewise relationship more closely than a simple quadratic one. This should not be interpreted as establishing the true structural form of the temperature-growth relationship, but it **illustrates how LASSO can choose among competing representations of related variables.**

Current GDP-per-capita growth receives the largest standardised LASSO coefficient at **2.5839**, indicating that recent economic performance is the strongest individual predictor in the model.

Other relatively large coefficients include CO_2 emissions per capita at 1.2206, population growth at -0.6794, heat stress at 0.6364 and CO_2 emissions per unit of GDP at 0.5879. These coefficients represent **conditional predictive associations** and should not be interpreted as estimates of causal economic or environmental effects.

Although LASSO retains almost all predictors, regularisation still reduces the magnitude of many coefficients relative to OLS. For example, the standardised temperature coefficient declines from -0.6226 under OLS to -0.4444 under LASSO, while the CO_2-per-capita coefficient declines from 1.3813 to 1.2206. Temperature squared is the only coefficient that is shrunk completely to zero. The exercise demonstrates that **LASSO can provide meaningful coefficient shrinkage even when it does not generate extensive feature elimination.**

OOS forecasting performance is relatively weak across all specifications. The historical-mean benchmark records the lowest RMSE at **5.1495**, compared with **5.2081** for OLS, **5.1715** for the Economic/Energy LASSO and **5.2004** for the Climate-Economy LASSO**.** None of the predictor-based models improves upon the historical-average benchmark according to RMSE.

The results are slightly different when assessed using MAE. The Economic/Energy LASSO achieves the lowest MAE at **3.3171**, compared with 3.3404 for the historical mean, 3.3452 for OLS and 3.3384 for the full Climate-Economy LASSO. I find some evidence that **economic and energy predictors modestly reduce typical forecast errors, but this improvement does not extend to RMSE because the models make substantial errors during extreme growth episodes.**

All models generate negative conventional out-of-sample (R^2) values. The historical mean, OLS, Economic/Energy LASSO and Climate-Economy LASSO produce values of **-0.1175, -0.1431, -0.1271 and -0.1398**, respectively. The benchmark-relative out-of-sample (R^2) values tell a similar story. OLS records **-0.0229**, the Economic/Energy LASSO records **-0.0086**, and the Climate-Economy LASSO records **-0.0199**. The Economic/Energy LASSO comes closest to matching the historical-mean benchmark, whereas none of the predictive specifications generates a positive out-of-sample (R^2).

One main objective of the exercise is to test whether adding direct climate information improves forecasting performance.

The Economic/Energy LASSO produces an RMSE of 5.1715, whereas the Climate-Economy LASSO produces an RMSE of 5.2004. Adding the climate variables changes RMSE by **-0.0289**, equivalent to a **-0.56% improvement**, meaning that forecast performance actually deteriorates slightly.

I consequently find that although LASSO selects three of the four climate variables, their inclusion does not provide incremental out-of-sample forecasting gains relative to the economic and energy specification.

This result highlights an important **distinction between variable selection and genuine predictive improvement.** The climate variables were retained because they provided useful information during cross-validation within the training period. However, this did not translate into superior performance during the final unseen test period. **A predictor can survive the LASSO penalty without necessarily improving forecast performance under a later economic regime.**

The coefficient-path figure provides further evidence that regularisation is relatively weak in this application. The selected alpha of 0.0070 lies near the low-penalty region of the path, where almost all coefficients remain non-zero. As alpha increases beyond this level, predictors progressively disappear and eventually even the dominant current-growth coefficient is eliminated.

The associated cross-validation-error curve reaches its minimum near the selected alpha and rises as stronger regularisation is imposed. The data favour retaining most of the available predictive information rather than imposing a highly sparse specification.

The yearly forecast comparison reveals an important source of the weak out-of-sample results. Forecasts are reasonably close to average realised growth during the earlier test years, but **all models fail dramatically around the COVID-19 shock.** Because predictors dated 2019 are used to forecast 2020 growth, the large contraction appears at 2019 in the forecast graph. Actual next-year average GDP-per-capita growth falls to approximately -5%, while all models continue forecasting positive growth.

The following predictor year, 2020, corresponds to the subsequent economic rebound, which the models again fail to anticipate. These observations generate exceptionally large errors and materially reduce overall RMSE and (R^2).

The failure around COVID-19 illustrates the difficulty of forecasting structural breaks using historical linear relationships. None of the predictors explicitly identifies the onset of a global pandemic, mandated economic shutdowns or the extraordinary policy responses that followed. Consequently, even a relatively rich climate-economy model remains unable to anticipate these extreme movements.

Forecast performance also differs across temperature regimes. For country-year observations with annual average temperatures at or below 13°C, the Climate-Economy LASSO achieves an RMSE of **4.7555**, compared with **5.4806** for observations above 13°C.

Forecast errors are approximately 15% larger in the warmer group. However, I do not interpret this difference causally. Countries with warmer climates also differ systematically in income, institutions, industrial composition and many other characteristics that may influence the volatility and predictability of economic growth.

The actual-versus-predicted scatter plot provides a final indication of the model's limitations. Predictions are concentrated mainly around moderate positive growth rates, whereas realised GDP-per-capita growth exhibits much wider variation, including severe contractions and large rebounds. The model displays substantial regression toward the mean and performs particularly poorly in the tails of the growth distribution.

(Refer to Appendix A.13 for more on Results)

Conclusion

Overall, the results show that climate variables contain some predictive information within the training sample, as 3 of the 4 direct climate indicators survive LASSO regularisation.

However, I find no evidence that their inclusion improves overall out-of-sample GDP-growth forecasting performance. The Economic/Energy LASSO marginally outperforms the Climate-Economy specification and comes closest to matching the historical-mean benchmark. The exercise demonstrates that variable selection, economic significance and out-of-sample predictive value are distinct concepts.

More broadly, the results illustrate an important **limitation of LASSO when predictors are highly correlated.** Temperature, temperature squared and heat stress are closely related, as are several of the emissions and energy-intensity variables. LASSO may favour one variable over another even when they contain overlapping information.

This limitation **provides a natural motivation for Elastic Net,** which combines the feature-selection properties of LASSO with the coefficient-stabilisation properties of Ridge Regression.

References & Bibliography

Stock, J.H. and Watson, M.W. (1999) 'Forecasting inflation', *Journal of Monetary Economics*, 44(2), pp. 293–335. doi: 10.1016/S0304-3932(99)00027-6. Stock and Watson article record

Tibshirani, R. (1996) 'Regression shrinkage and selection via the Lasso', *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1), pp. 267–288. doi: 10.1111/j.2517-6161.1996.tb02080.x. Tibshirani article

Federal Reserve Bank of St. Louis (2026) *Federal Reserve Economic Data (FRED)*. St. Louis, MO: Federal Reserve Bank of St. Louis. Available at: FRED Economic Data (Accessed: 22 August 2026).

Welch, I. and Goyal, A. (2008) 'A comprehensive look at the empirical performance of equity premium prediction', *The Review of Financial Studies*, 21(4), pp. 1455–1508. doi: 10.1093/rfs/hhm014.

Goyal, A. (2026) *Updated data for Goyal and Welch (2008): U.S. equity premium predictor data*. University of Lausanne and Swiss Finance Institute. Available at: Amit Goyal data library (Accessed: 22 August 2026). The site provides the updated Goyal–Welch dataset through 2025.

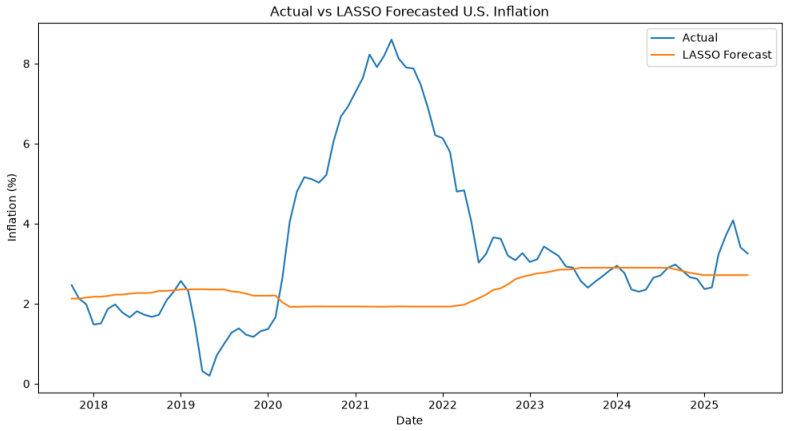
Burke, M., Hsiang, S.M. and Miguel, E. (2015) 'Global non-linear effect of temperature on economic production', *Nature*, 527, pp. 235–239. doi: 10.1038/nature15725.

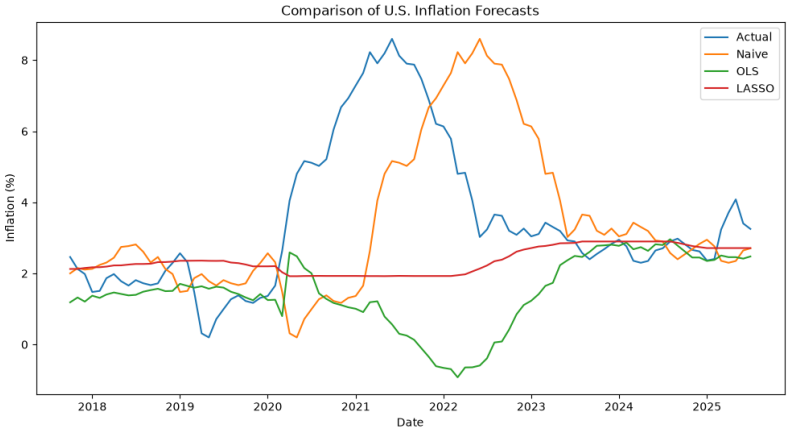
Our World in Data (2026) *CO_2 and Greenhouse Gas Emissions dataset*. Our World in Data. Available at: OWID CO_2 dataset (Accessed: 23 August 2026). The dataset contains annual country-level emissions, energy, population and GDP-related variables used in this exercise.

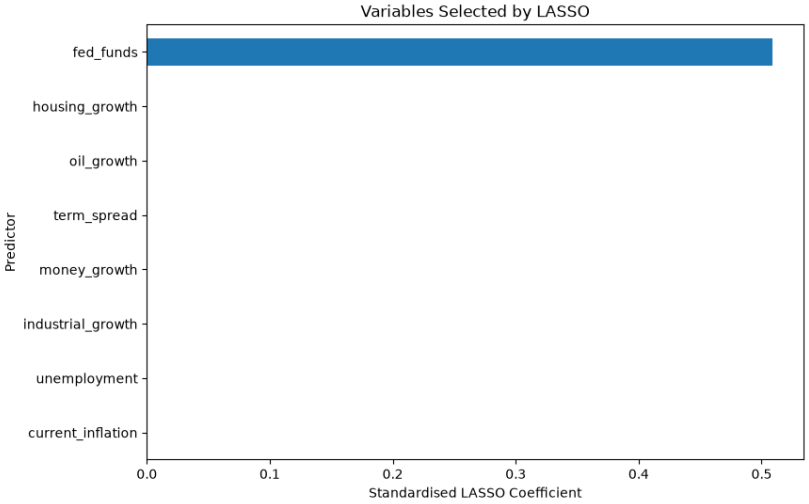
Copernicus Climate Change Service (2026), with processing by Our World in Data *Average annual surface temperature* [dataset]. Based on ERA5 reanalysis data. Available at: OWID annual surface temperature dataset (Accessed: 23 August 2026).

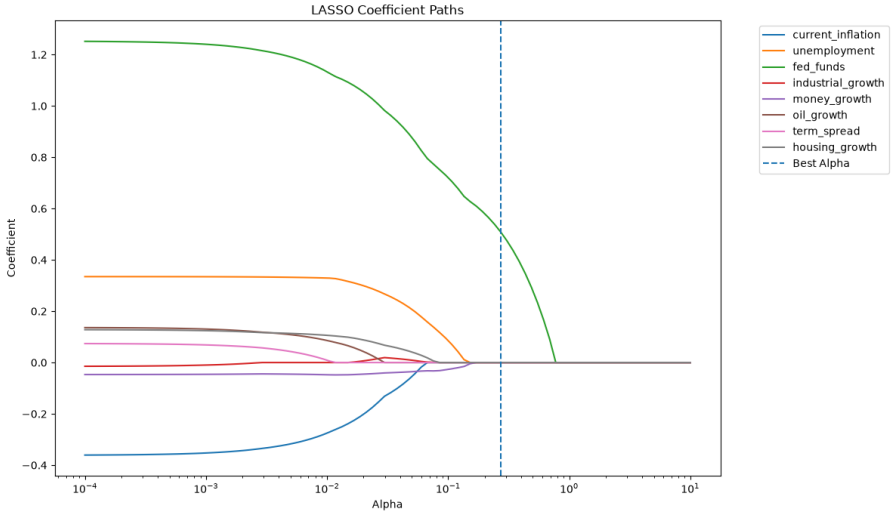
Appendices

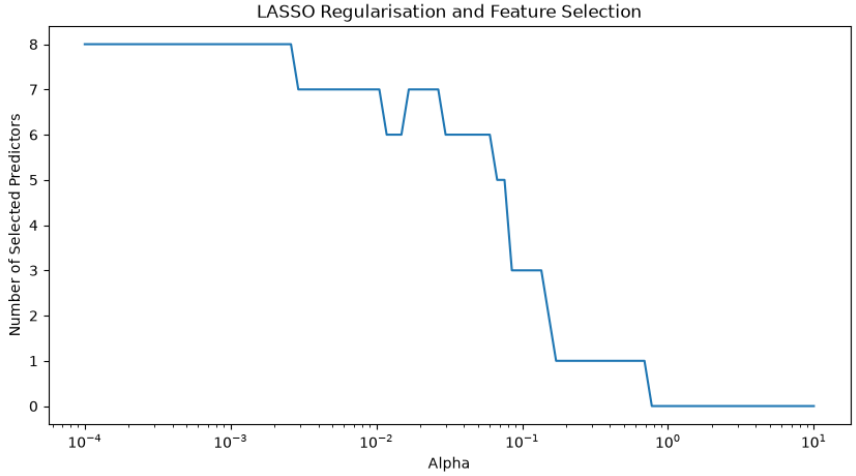
A.11 Forecasting U.S. Inflation Using Macroeconomic Indicators with LASSO

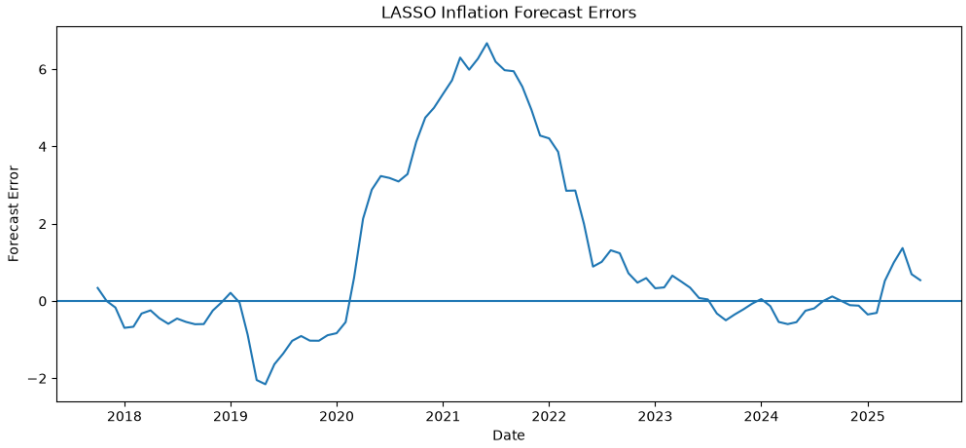







LASSO Coefficient Paths
current_inflation
unemployment
fed_funds
industrial_growth
money_growth
oil_growth
term_spread
housing_growth
Best Alpha
Coefficient
Alpha


LASSO Regularisation and Feature Selection
Number of Selected Predictors
Alpha

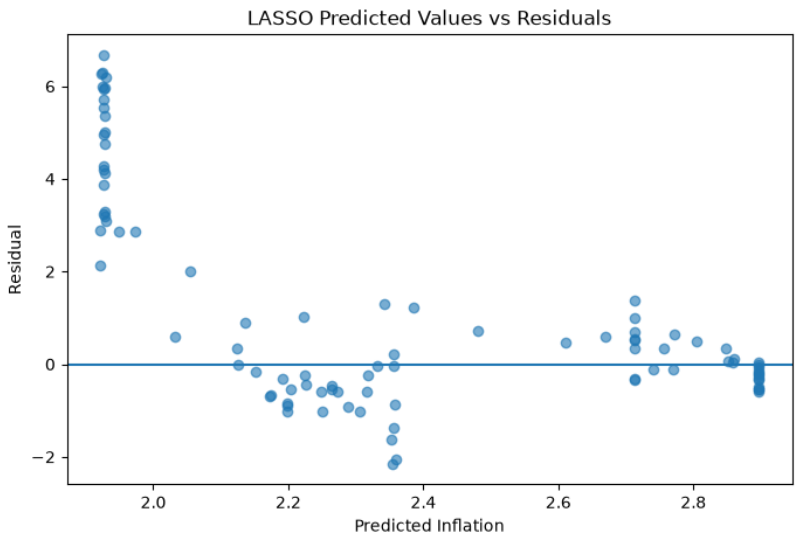

LASSO Inflation Forecast Errors
Forecast Error
Date

	OLS	LASSO	Selected_by_LASSO
current_inflation	-0.3615	0.0000	False
unemployment	0.3348	0.0000	False
fed_funds	1.2530	0.5085	True
industrial_growth	-0.0151	0.0000	False
money_growth	-0.0469	-0.0000	False
oil_growth	0.1368	0.0000	False
term_spread	0.0744	-0.0000	False
housing_growth	0.1283	0.0000	False

Figure 1: *Final Coefficient Table for US Inflation*

	Model	RMSE	MAE	R2
0	Naive	2.5095	1.8298	-0.4437
1	OLS	3.3681	2.1979	-1.6007
2	LASSO	2.5638	1.6691	-0.5070

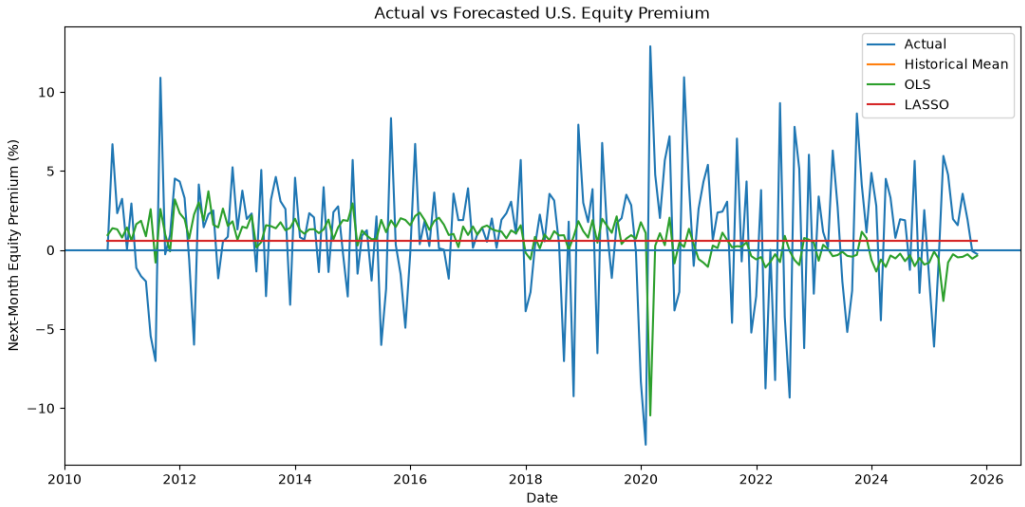
Figure 2: *Comparative Performance of Models for US Inflation*

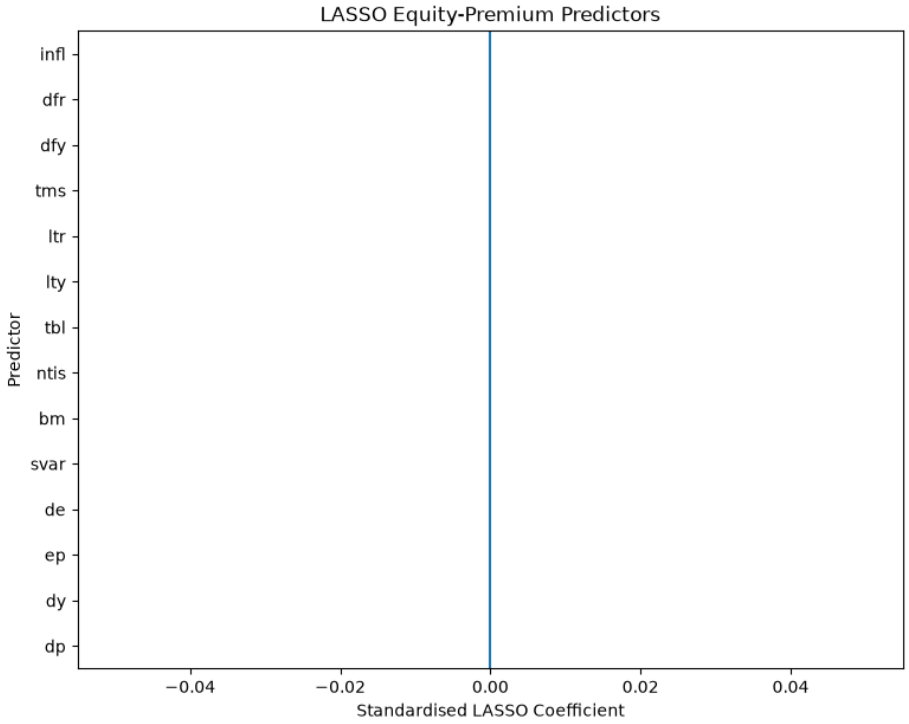


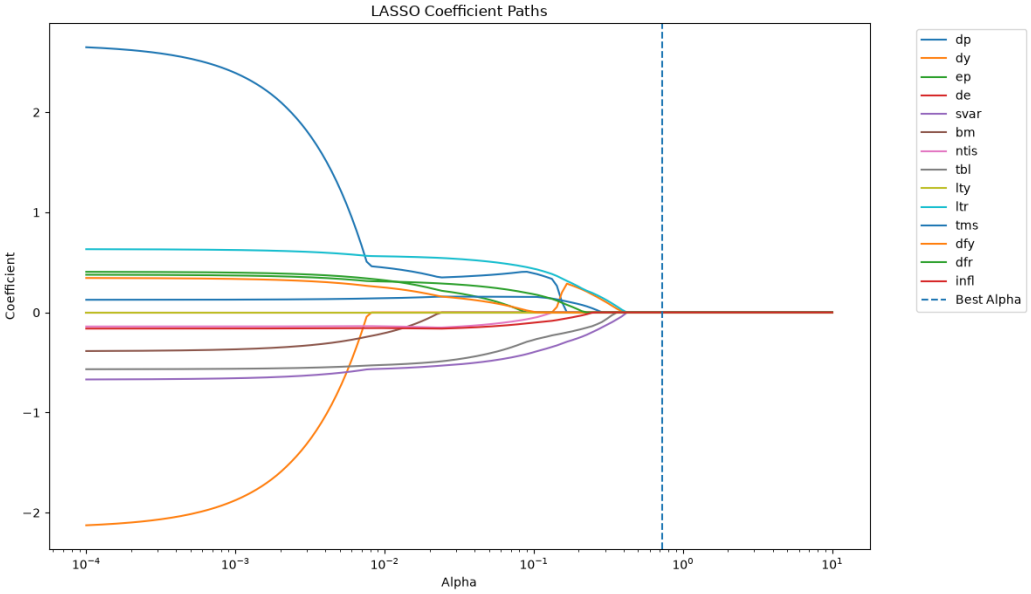
A.12 Predicting the U.S. Equity Premium Using Economic and Financial Predictors

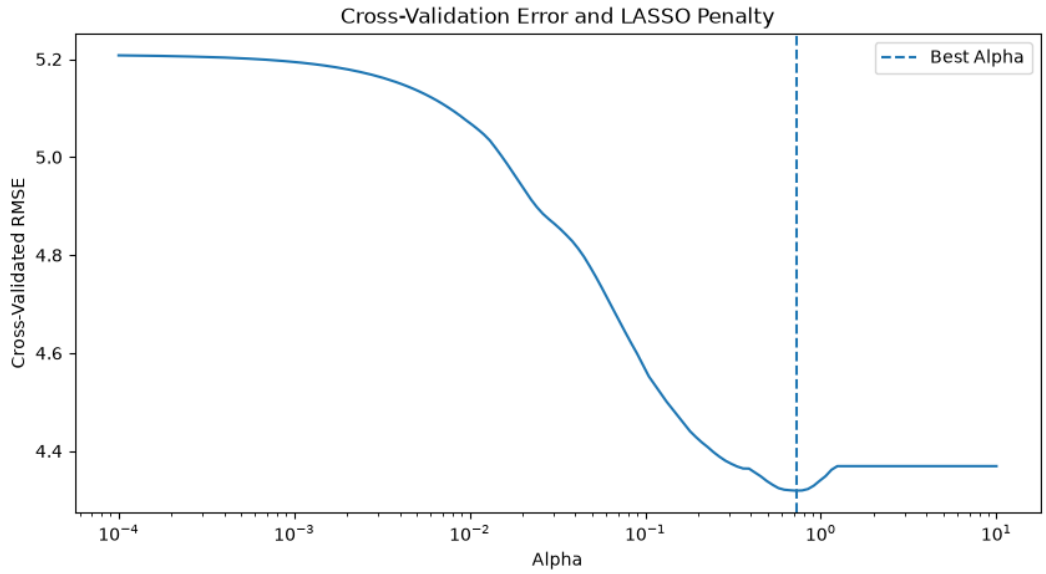
	OLS	LASSO	Selected_by_LASSO
dp	1.8110	0.0	False
dy	-2.1541	0.0	False
ep	1.3253	0.0	False
de	0.5933	0.0	False
svar	-0.6707	0.0	False
bm	-0.3876	0.0	False
ntis	-0.1406	0.0	False
tbl	-0.3251	0.0	False
lty	-0.2248	0.0	False
ltr	0.6326	0.0	False
tms	0.2430	0.0	False
dfy	0.3451	0.0	False
dfr	0.3774	0.0	False
infl	-0.1615	0.0	False

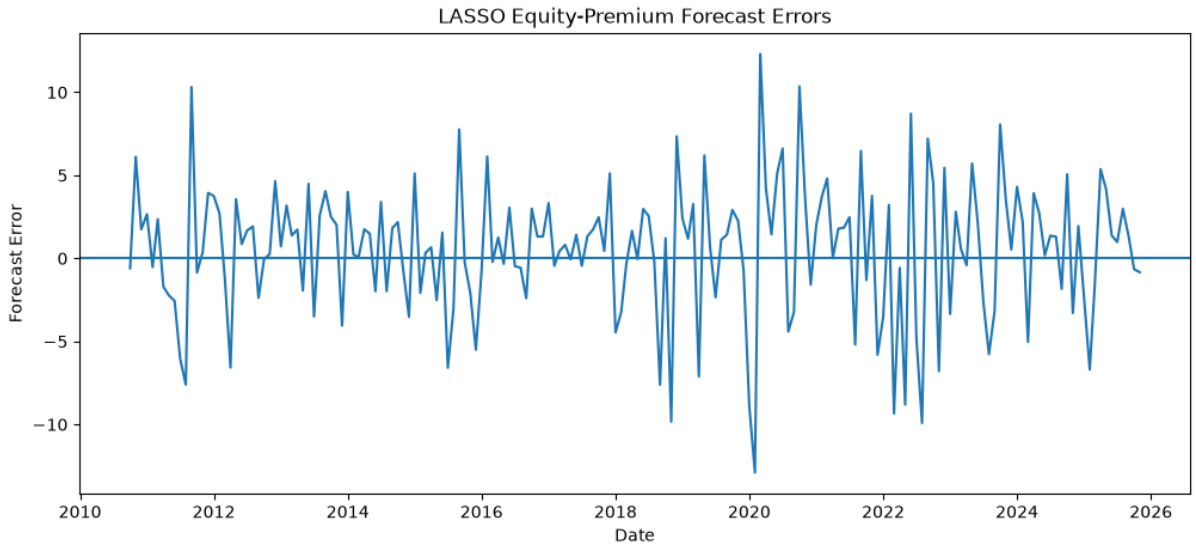
Figure 3: *Coefficient Table for US Equity Premium*

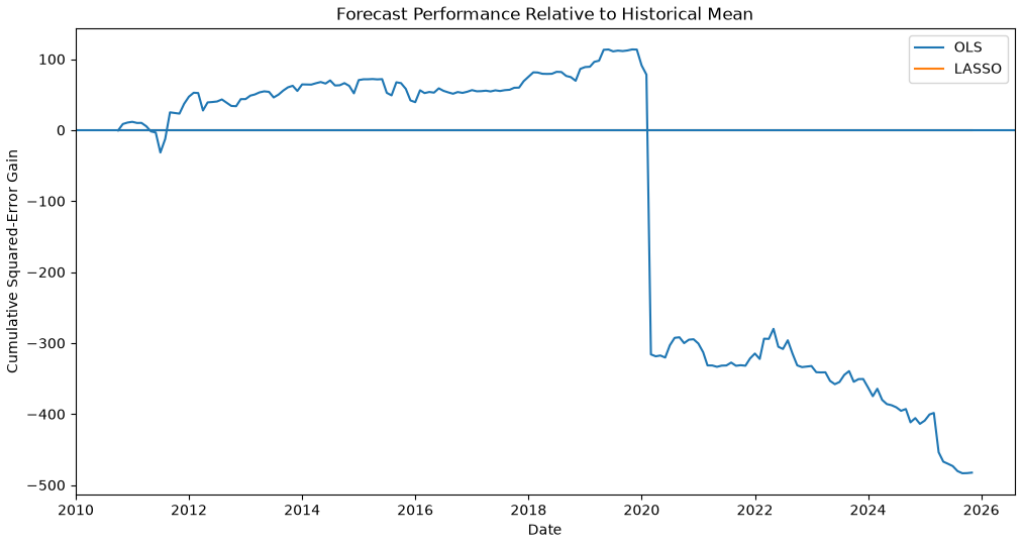



LASSO Equity-Premium Predictors
infl
dfr
dfy
tms
ltr
lty
tbl
ntis
bm
svar
de
ep
dy
dp
Predictor
−0.04
−0.02
0.00
0.02
0.04
Standardised LASSO Coefficient

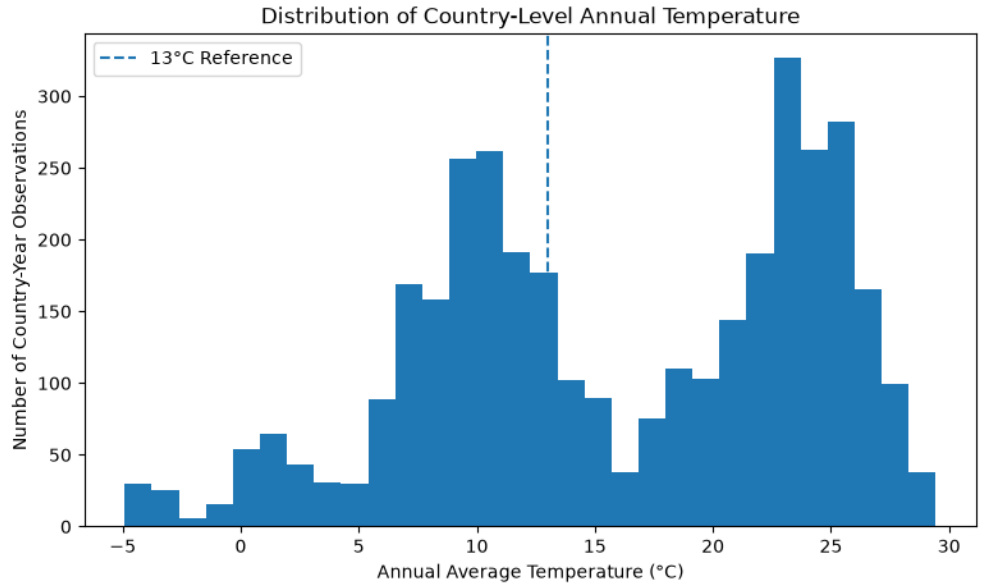

LASSO Coefficient Paths
Coefficient
Alpha
dp
dy
ep
de
svar
bm
ntis
tbl
lty
ltr
tms
dfy
dfr
infl
Best Alpha

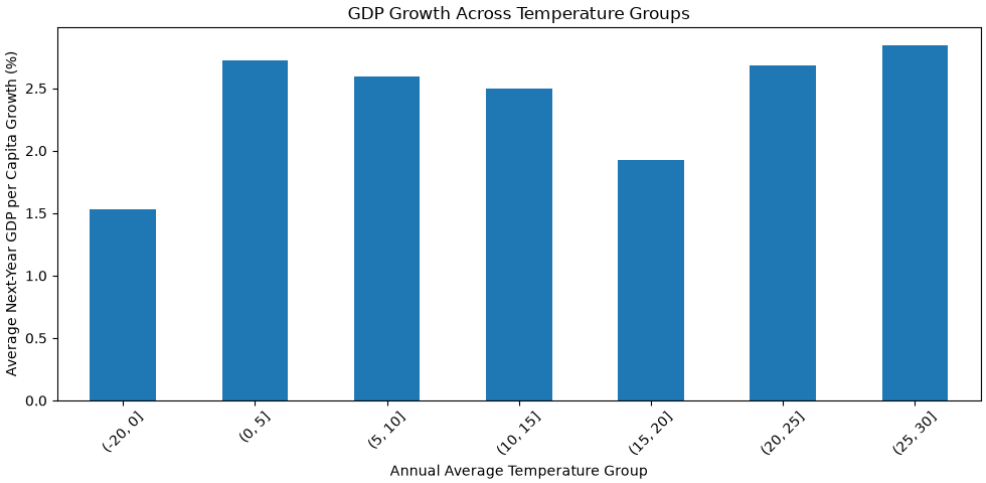

Cross-Validation Error and LASSO Penalty
Best Alpha
Cross-Validated RMSE
5.2
5.0
4.8
4.6
4.4
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
Alpha


LASSO Equity-Premium Forecast Errors
Forecast Error
10
5
0
−5
−10
2010
2012
2014
2016
2018
2020
2022
2024
2026
Date


Forecast Performance Relative to Historical Mean
OLS
LASSO
Cumulative Squared-Error Gain
100
0
−100
−200
−300
−400
−500
2010
2012
2014
2016
2018
2020
2022
2024
2026
Date

A. 13 Predicting GDP Growth Under Climate Stress Using Economic, Energy and Temperature Indicators



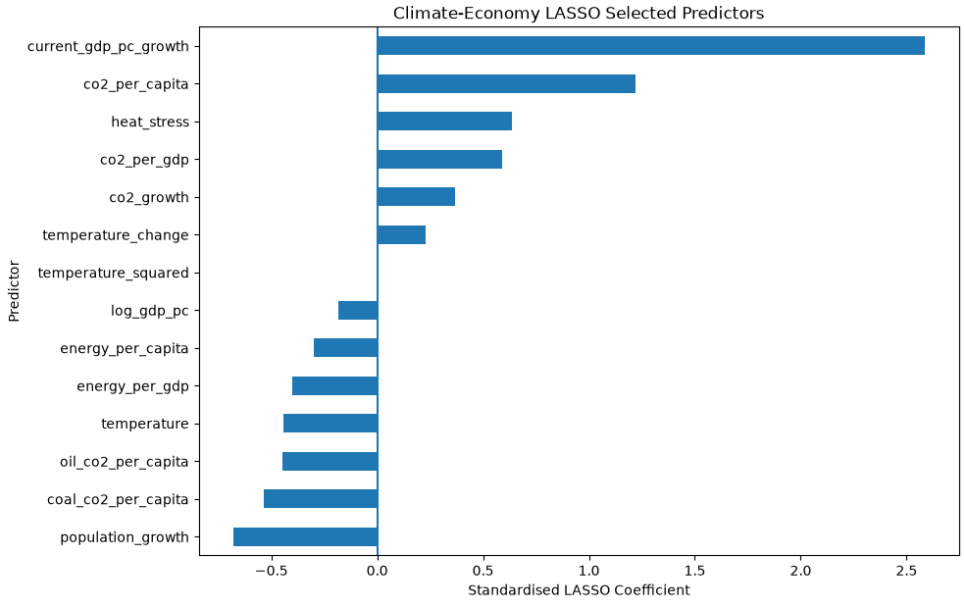


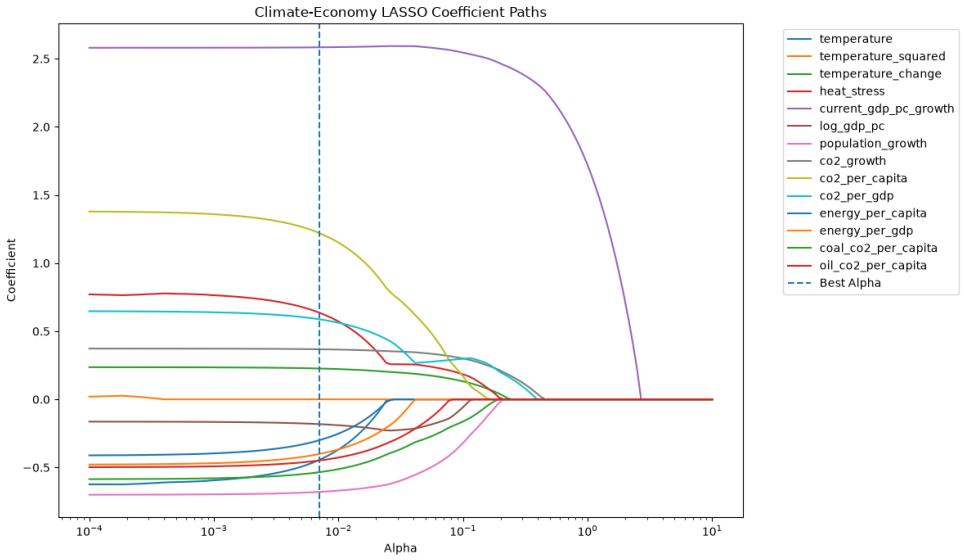
	OLS	LASSO	Selected_by_LASSO	Variable_Group
temperature	-0.6226	-0.4444	True	Climate
temperature_squared	0.0105	0.0000	False	Climate
temperature_change	0.2361	0.2265	True	Climate
heat_stress	0.7783	0.6364	True	Climate
current_gdp_pc_growth	2.5805	2.5839	True	Economic/Energy
log_gdp_pc	-0.1634	-0.1818	True	Economic/Energy
population_growth	-0.7007	-0.6794	True	Economic/Energy
co2_growth	0.3729	0.3676	True	Economic/Energy
co2_per_capita	1.3813	1.2206	True	Economic/Energy
co2_per_gdp	0.6477	0.5879	True	Economic/Energy
energy_per_capita	-0.4124	-0.3002	True	Economic/Energy
energy_per_gdp	-0.4798	-0.4012	True	Economic/Energy
coal_co2_per_capita	-0.5866	-0.5349	True	Economic/Energy
oil_co2_per_capita	-0.4986	-0.4489	True	Economic/Energy

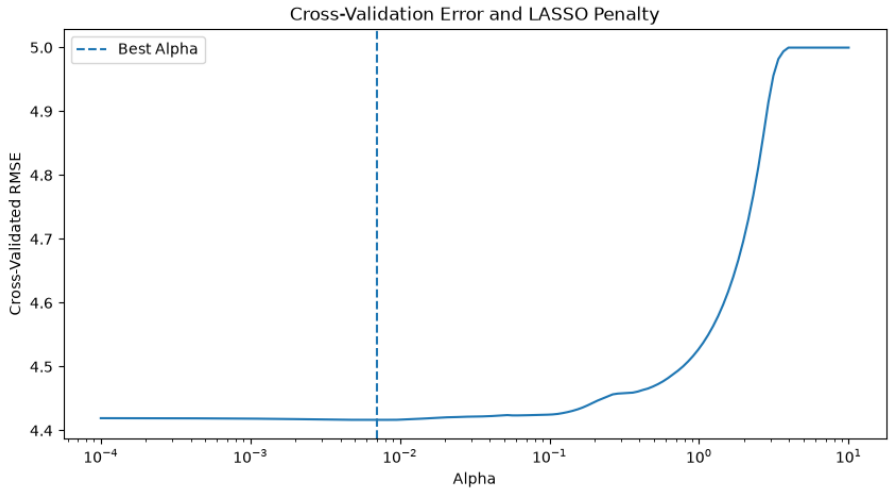
Figure 4: *Selection of Predictors – OLS versus LASSO*

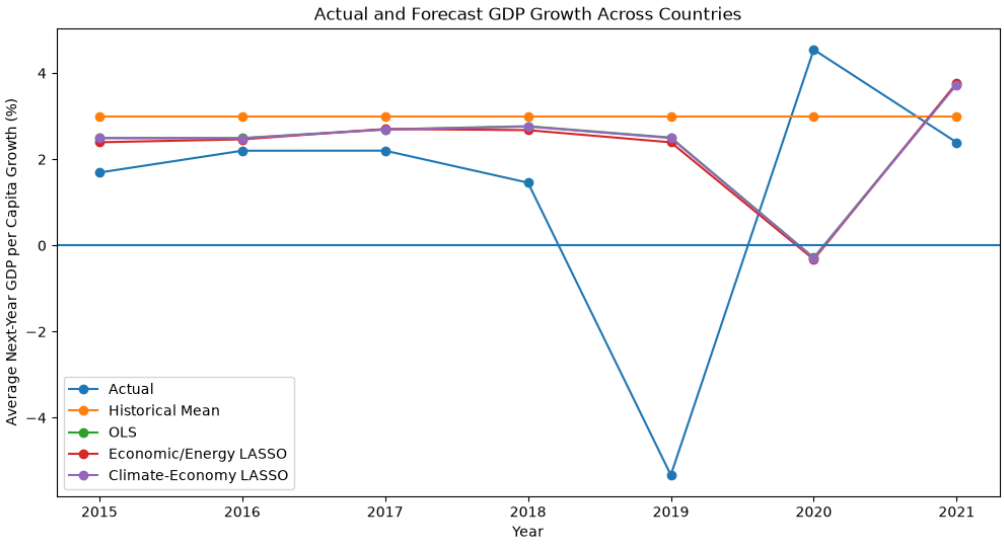
	Model	RMSE	MAE	R2
0	Historical Mean	5.1495	3.3404	-0.1175
1	OLS	5.2081	3.3452	-0.1431
2	Economic/Energy LASSO	5.1715	3.3171	-0.1271
3	Climate-Economy LASSO	5.2004	3.3384	-0.1398

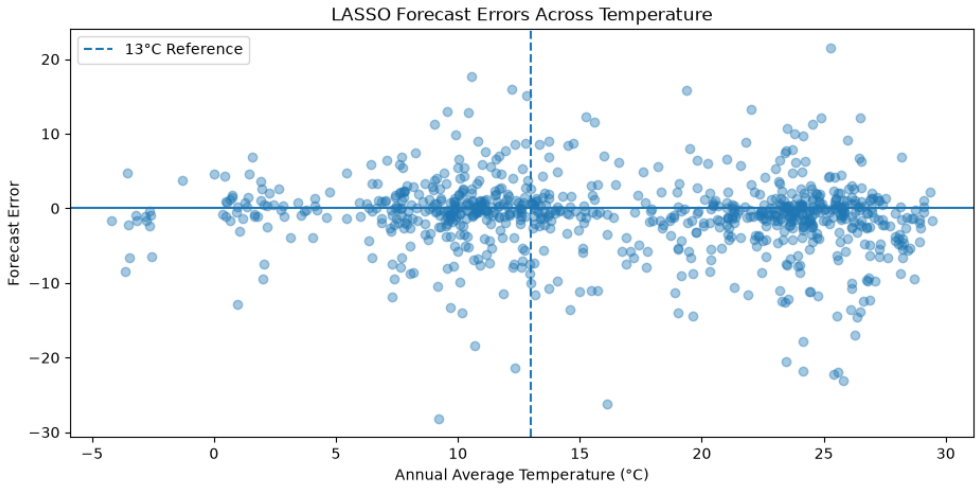
Figure 5: *Performance Table for GDP Growth under Climate Stress*

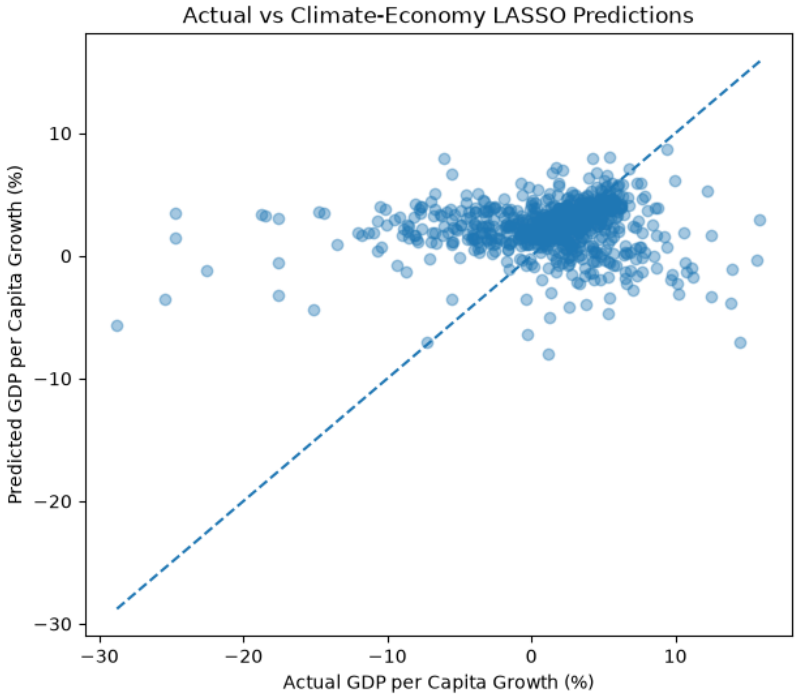

Climate-Economy LASSO Selected Predictors
current_gdp_pc_growth
co2_per_capita
heat_stress
co2_per_gdp
co2_growth
temperature_change
temperature_squared
log_gdp_pc
energy_per_capita
energy_per_gdp
temperature
oil_co2_per_capita
coal_co2_per_capita
population_growth
Predictor
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
Standardised LASSO Coefficient


Climate-Economy LASSO Coefficient Paths
Coefficient
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
Alpha
temperature
temperature_squared
temperature_change
heat_stress
current_gdp_pc_growth
log_gdp_pc
population_growth
co2_growth
co2_per_capita
co2_per_gdp
energy_per_capita
energy_per_gdp
coal_co2_per_capita
oil_co2_per_capita
Best Alpha


Cross-Validation Error and LASSO Penalty
Best Alpha
Cross-Validated RMSE
5.0
4.9
4.8
4.7
4.6
4.5
4.4
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
Alpha


Actual and Forecast GDP Growth Across Countries
Average Next-Year GDP per Capita Growth (%)
Year
Actual
Historical Mean
OLS
Economic/Energy LASSO
Climate-Economy LASSO


LASSO Forecast Errors Across Temperature
13°C Reference
Forecast Error
Annual Average Temperature (°C)


Actual vs Climate-Economy LASSO Predictions
Predicted GDP per Capita Growth (%)
Actual GDP per Capita Growth (%)